Performance Evaluation and Optimization of HBM-Enabled GPU for Data-Intensive Applications

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Summary of paper:
M. Zhu, Y. Zhuo, C. Wang, W. Chen, and Y. Xie, “Performance evaluation and optimization of HBM-Enabled GPU for data-intensive applications,” IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 2018.

Key lessons to be learnt from this lecture

- Comparison of GPUs with different capabilities and main memory of different types/bandwidth
  - AMD Fury X
  - NVIDIA K40
  - AMD FirePro W7100
- Impact of these features on memory management during DNN training
- Comparison of CUDA Caffe vs OpenCL Caffe
- During DNN training, transferring data from GPU memory to host memory

Bandwidth values in different systems

- Of Cerebras WSE-1 and WSE-2 (on-chip) = 9PB/s and 20 PB/s
- on using persistent storage in FPGA in Brainwave (on-chip)= 35840 GB/s
- Of Volta GPU (uses 4 HBM2) = 900 GB/s
- Of TPU v2 and v3 (uses HBM2)= 700 and 900 GB/s
- of AMD Fury X (uses HBM)= 512 GB/s
- of MCDRAM in Phi (uses HMC)= 500 GB/s
- of K40 GPU (uses GDDR5)= 288 GB/s
- Of KNM Phi (uses DDR4) = 115 GB/s
- of NVLINK = 80 GB/s
- of E5-2697 v2 CPU (uses DDR3) = 59.7 GB/s
- of TPUv1 (uses DDR3) = 34 GB/s
- Highest PCIe bandwidth = 16 GB/s

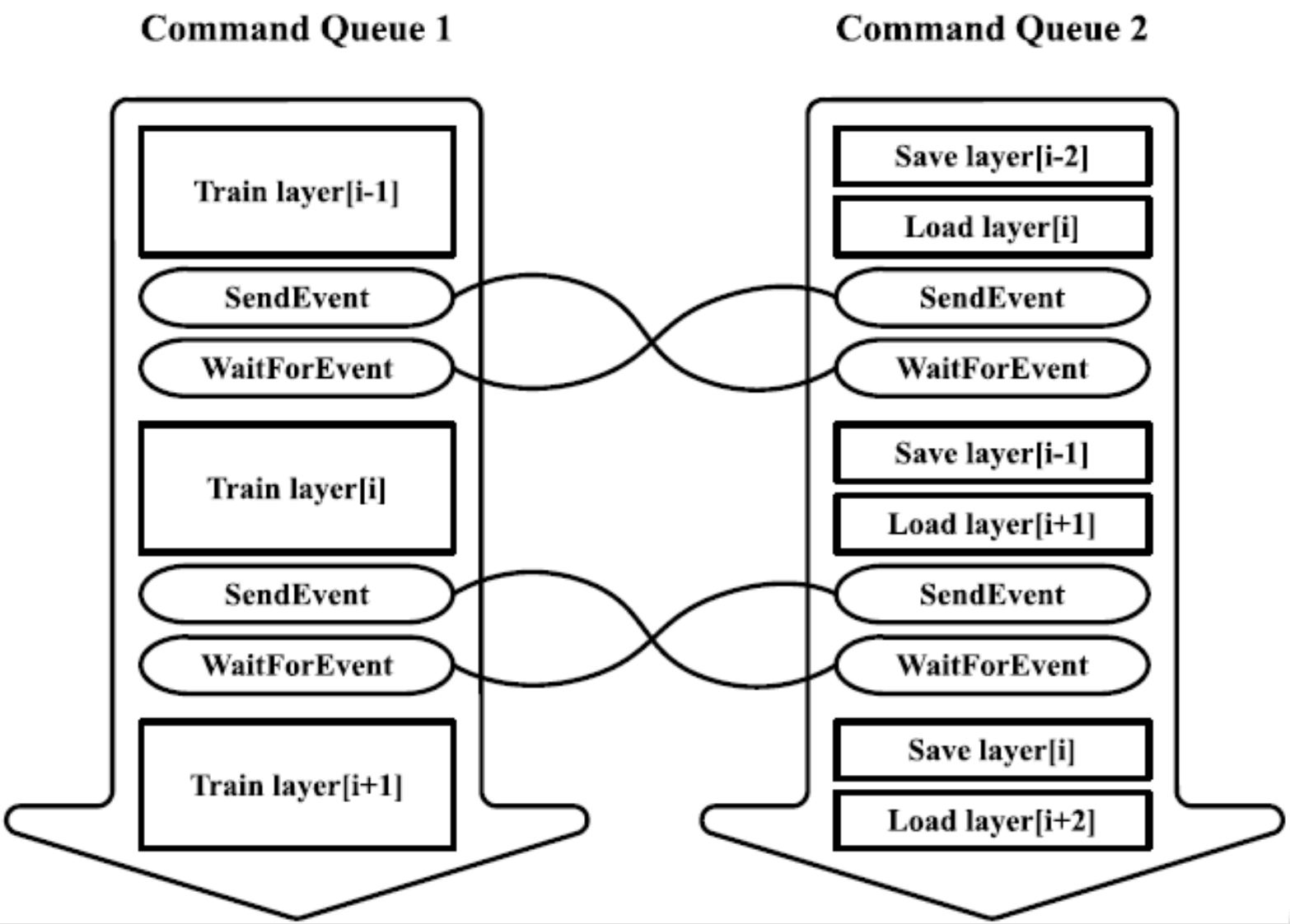
Motivation

- On using minibatch-size of 256, training of AlexNet becomes very slow on the HBM-enabled GPU
- Reason: feature maps cannot be stored in HBM due to its limited capacity
- When GPU memory becomes insufficient, OpenCL runtime allocates memory buffers in the host memory.
- Afterwards, read/write operations are performed to this buffer over PCIe, which provides much lower bandwidth than HBM.

Proposed Technique

- Due to the layer-wise training, in Backward propagation phase, only the output of one layer is required when the next layer is being processed.
- Their technique offloads data of layers that are required in future to CPU memory and prefetches their data before they are required.
- Training of a CONV layer takes larger time than the data-transfer between CPU and GPU
- ➔the data-transfer latency is easily hidden.

Implementation



Implementation

- Their technique works using two command queues.
- At the time of training layer K,
  - the data of layer K-1 is copied from device to host memory and
  - that of layer K+1 is copied from host to device memory.
- When the data of layer K+1 arrives, the second queue notifies first queue to begin training layer K+1.
- When layer K+1 has been processed, the first queue notifies the second queue to begin data-transfer.

GPU Specification

- AMD FirePro W7100 has same architecture as Fury X GPU but it uses GDDR5 as the off-chip memory. W7100 can be seen as a Fury X GPU without the HBM
- NVIDIA Tesla K40 is latest high-end GPU for general purpose computing

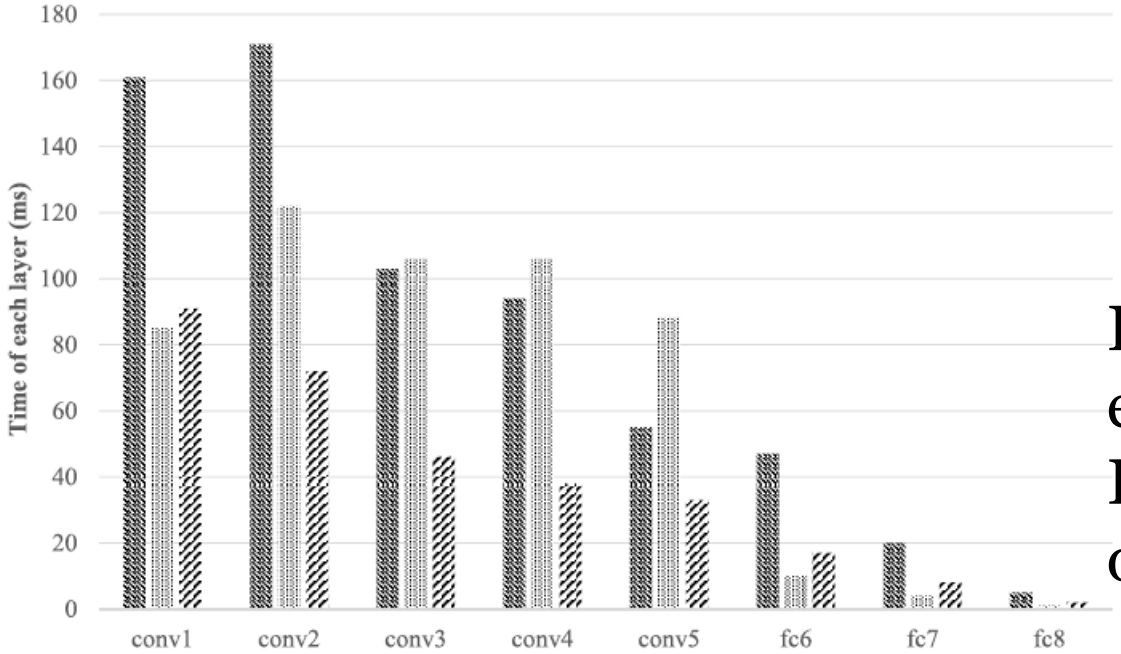
	AMD W7100	NVIDIA K40	AMD Fury X
Number of Compute Units	28	15	64
Number of Cores	1792	2880	4096
Architecture	GCN	Kepler	GCN
Memory Size(GB)	8	12	4
Memory Bus Width(bit)	256	384	4096
Peak Bandwidth(GB/s)	160	288	512

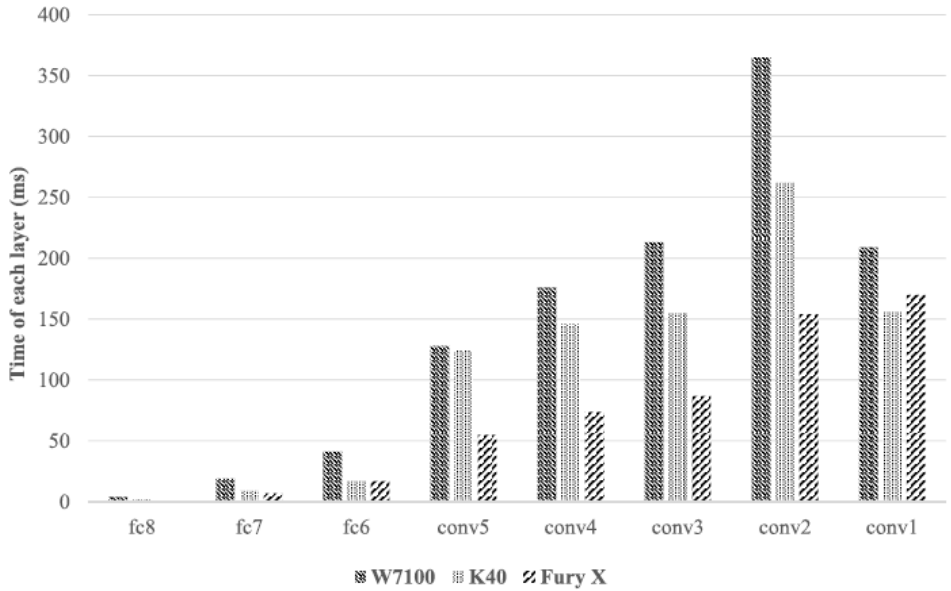
Comparison

- Fury X has more than 2X compute units than AMD W7100 GPU.
- Still it has no memory congestion problem because peak memory bandwidth of the Fury X is more than 3X that of W7100.
- Limitation of Fury X is that its HBM has low capacity.

- Experiments: OpenCL Caffe used on AMD GPUs and CUDA Caffe on NVIDIA GPUs

Layer-wise execution time of AlexNet training





For each layer, execution time on Fury X is half of time on W7100.

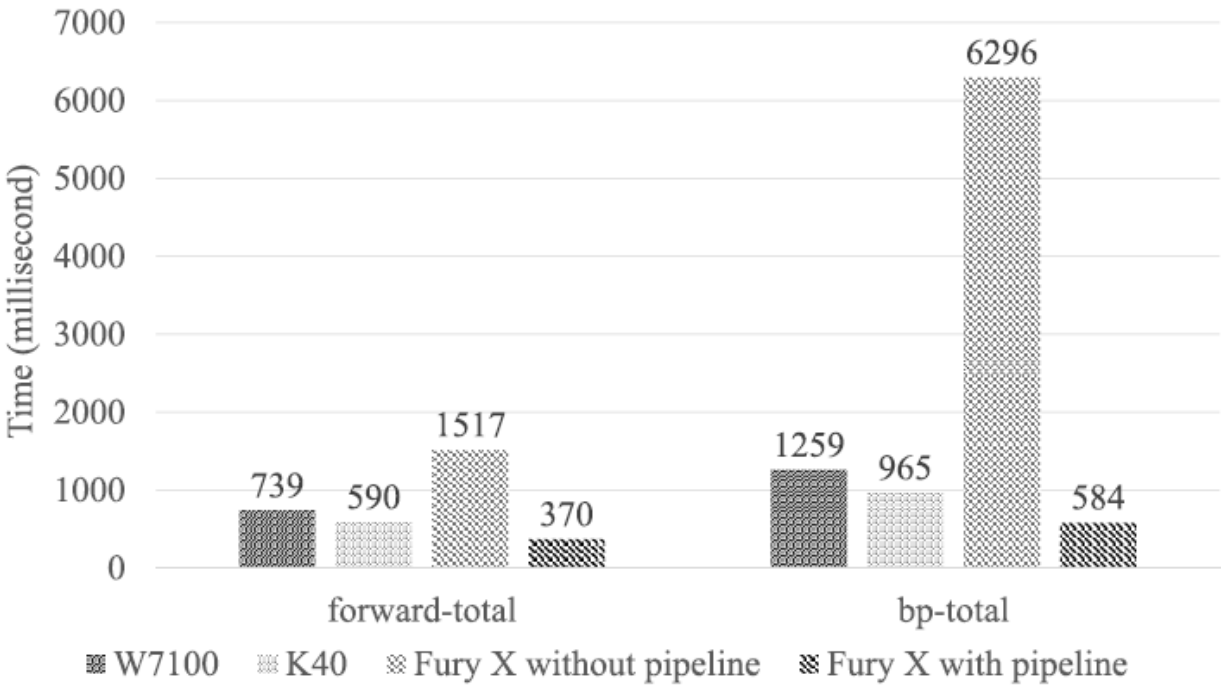
=> HBM-enabled GPU can achieve significant speedup without any code-change.

Comparing Fury X with K40

- Comparing Fury X with K40, winner varies for different layers.
- Reason1: Difference in architectures. K40 integrates more processing elements and more complex scheduling in its compute units which reduces the total number of compute units.
- Reason2: CUDA Caffe calls cuBLAS for linear algebra computation. OpenCL Caffe uses clBLAS.

Execution time of one iteration

- Although Fury X outperforms K40 in the processing of some layers, its training speed for one iteration is much slower than the other two GPUs



Reason for poor performance of Fury X

- AlexNet training requires more than 4 GB memory.
- OpenCL automatically allocates memory buffer objects on CPU (host) memory when system is out of GPU (device) memory.
- => Compute units have to stall to fetch data from CPU-end memory objects => GPU compute units stay idle.
- For HBM-enabled memory, situation is even worse because large number of compute units on chip will cause severe PCIe congestion.

Applying their technique

- They apply their proposed technique to all convolution layers.
- Results: total data requirement is reduced from 4.35 to 3.52 GB.
- After reducing global memory usage, Fury X outperforms K40 with a speedup of 1.63X.